

Thomson Tong

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EDUCATION

University of Waterloo

BASc Biomedical Engineering

Sep 2024 – Present

- C++ Programming, Data Structures & Algorithms, Software Design (BME121/122)
- Agile Development, Design Patterns, Version Control, Development Cycles (BME261)
- Probability & Statistics, Experimental Design (BME213)

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, Rust

Frameworks & Libraries: React, Node.js, Express, FastAPI

Databases & Infrastructure: PostgreSQL, MongoDB, Docker, Linux, Git, CI/CD (GitHub Actions)

EXPERIENCE

Undergraduate Research Assistant | University of Waterloo VIP Lab | Waterloo, ON

May 2026 – Present

- Building a modular data pipeline that processes egocentric video to extract and estimate food consumption metrics, integrating multiple image processing and 3D reconstruction stages into an end-to-end system.
- Developing a multi-stage object isolation and tracking workflow for video streams, orchestrating multiple model inferences within a unified processing architecture.
- Implementing automated benchmarking and validation tooling to continuously measure pipeline output accuracy against reference datasets across multiple evaluation metrics.

Full Stack Developer (Co-op) | MissionPerform | Mississauga, ON

Jan 2026 – Apr 2026

- Resolved **30+** high-priority production tickets using the **MERN** stack, improving system uptime and reducing technical debt for a B2B pharmaceutical manufacturing SaaS platform.
- Shipped a high-priority feature within a **1-week sprint** as part of a 3-person team, using AI-assisted development workflows to accelerate iteration.
- Improved a cross-platform video-conferencing system for tracking factory workspace issues by implementing user invitations, automated email notifications, and scheduled calling workflows, boosting connection stability by **25%**.

AI Research Member | Wat.AI | Waterloo, ON

Jun 2025 – Dec 2025

- Co-authored a research report (arXiv:2512.23028) analyzing model architectures and tokenization behaviors to establish system constraints for a video-to-artifact processing pipeline.

Systems Diagnostics Technician | M-Zone Auto | Markham, ON

Jun 2025 – Aug 2025

- Performed root cause analysis, systems testing, and fault isolation across **250+** vehicles, applying structured debugging methodologies to identify and resolve data-routing failures.

PROJECTS

Transight | Full-Stack Transit Predictive Visualization System

Python, FastAPI, LightGBM, Mapbox

- Developed a full-stack transit analytics web application using Python-based data pipelines and predictive modeling, achieving a **50% improvement** in prediction accuracy over baseline approaches.
- Architected asynchronous REST API services with FastAPI, delivering real-time data under **sub-100ms latency**.

RL Poker Bot | Self-Play Reinforcement Learning Engine for 6-Max NL Hold'em

Python, PyTorch, NumPy, TensorBoard

- Designed and implemented an object-oriented simulation framework in Python, synchronizing game state across **1,000** parallel execution environments for large-scale self-play training.
- Improved backend training performance by **25%** through runtime optimization and automated validation, monitoring execution health across 500+ training cycles and 4 testing baselines.

ECG-Analysis | Multi-label Signal Classification Pipeline

Python, NumPy, Pandas, scikit-learn

- Built a multi-label signal classification pipeline achieving **94% accuracy** across 4 diagnostic categories.
- Engineered a structured data ingestion and preprocessing pipeline for biomedical datasets, reducing processing latency by **20%**.